

Prehistoric VR Simulations as educational content

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Declaration of the candidate & Supervisor

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the Co supervisor

Date:

Abstract

The *Prehistoric Animal Behaviour Simulation* presents a virtual reality–based approach for modelling lifelike behaviour of extinct species through autonomous artificial intelligence. Modern VR applications often rely on scripted or state-machine driven animal actions, which limits realism and reduces educational value. This study introduces a machine-learning-enhanced behaviour system implemented within Unity, designed to simulate dynamic responses such as aggression, avoidance, herding, territoriality, and environmental awareness. By integrating ML-Agents with custom behavioural parameters—confidence level, health state, stimulus sensitivity, and inter-species interaction rules—the system enables animals to react adaptively to both users and surrounding entities.

The methodology consists of analysing existing behaviour-modelling techniques, designing behaviour architectures suitable for real-time VR environments, training agents in controlled scenarios, and deploying them in an immersive ecosystem optimized for Meta Quest hardware. The simulation aims to improve user engagement and support experiential learning in contexts such as museums, classrooms, and edutainment environments. A feasibility study was conducted across technical, financial, operational, and ethical dimensions, supported by testing results that evaluate performance, behaviour consistency, VR usability, and responsiveness under varying environmental conditions.

The resulting system demonstrates that autonomous, learning-based prehistoric animals significantly enhance immersion by moving beyond predictable scripted patterns. User observations indicated improved realism, higher engagement, and a more convincing portrayal of extinct ecosystems. Findings also highlight potential for further expansion through multi-species ecosystems, curriculum-aligned educational modules, and wider commercialization in educational VR markets. Overall, the project provides a scalable foundation for scientifically inspired, interactive prehistoric simulations that bridge entertainment and learning through AI-driven behaviour models.

Keywords:

Virtual Reality

Machine Learning Agents

Autonomous Behaviour

Prehistoric Simulation

Unity

1.Introduction

1.1 Background and Literature Review

Virtual Reality (VR) has rapidly evolved into a powerful medium for experiential learning, scientific visualization, and immersive entertainment. Modern VR platforms enable users to explore environments that are otherwise inaccessible—whether due to danger, cost, distance, or historical extinction. Among these domains, prehistoric worlds have gained significant attention due to their cultural, educational, and scientific appeal. While VR technology has become increasingly capable of reproducing lifelike environments, a critical limitation persists: the behaviour of virtual animals is often static, predictable, or heavily scripted, reducing the authenticity of the experience and hindering educational value.

Prehistoric animals, unlike contemporary species, cannot be directly observed or behaviourally studied. Therefore, simulations depend on reconstructed behavioural models derived from biological research, fossil evidence, and comparative studies of modern animal analogues. Traditional VR-based wildlife simulations typically adopt finite-state machines, behaviour trees, or simple rule-based AI to control animal motion and reactions. These approaches, despite being functional, tend to produce repetitive patterns and lack adaptive decision-making. As a result, the animals in these simulations appear artificial and fail to reflect the complexity of real ecological interactions such as territorial dynamics, herding behaviour, predator–prey relationships, and environmental responsiveness.

Recent work has attempted to address this limitation through artificial intelligence. Turan and Çetin (2019) introduced the use of fuzzy-logic-based behaviour models in a virtual island ecosystem, allowing virtual animals to make decisions based on confidence and health parameters rather than fixed scripts. Their contribution demonstrated that even simple AI techniques can produce more believable behaviours compared to traditional methods. Similarly, Naik et al. (2020) explored how animals interact within virtual environments from an experimental perspective, highlighting the importance of sensory realism, dynamic stimulus responses, and feedback loops. Although their focus is on real animals placed inside virtual settings, their conceptual framework offers foundational insights into modelling behaviour within artificial ecosystems.

More recently, VR systems have been integrated into museum exhibitions, as demonstrated in DinosaurVR by Carvalho Souza (2023), which presented prehistoric content through immersive environments designed for public education. However, such systems primarily focused on visual fidelity and user experience rather than autonomous animal behaviour. The simulations used in these exhibitions remain limited in behavioural complexity and still depend on predetermined animations and scripted interactions.

This literature collectively highlights a significant gap: VR-based prehistoric simulations rarely feature autonomous, self-adjusting animal behaviour that adapts to environmental stimuli and user interactions. Existing works either emphasize visual immersion without behavioural realism or apply AI techniques that are insufficiently dynamic for VR ecosystems. These constraints reduce the effectiveness of VR as a tool for learning and scientific engagement.

The *Prehistoric Animal Behaviour Simulation* project aims to bridge this gap by incorporating machine learning–driven autonomous behaviours into a VR ecosystem developed in Unity. Through the use of ML-Agents, custom behavioural attributes, and ecologically inspired interaction rules, the system aspires to create lifelike prehistoric animals capable of reacting intelligently to both players and environmental factors. This aligns with global movements toward more naturalistic AI agents in virtual environments, supporting higher realism, deeper educational impact, and broader usability in museums, schools, and entertainment platforms.

1.2 Research Gap

Existing virtual reality experiences that feature prehistoric environments predominantly prioritize visual immersion, environmental artistry, and user interaction, while the behaviour of virtual animals remains largely simplified. Most commercially available and academic VR simulations depend on predefined animations, behaviour trees, or fixed state machines, which restrict animals to predictable and repetitive actions. These systems lack the adaptive decision-making required to portray lifelike ecological dynamics such as herding, foraging, predator–prey interactions, territorial displays, and reactive movement in response to player presence.

Although prior research has demonstrated incremental improvements in behavioural realism through rule-based AI and fuzzy-logic decision systems, these approaches do not scale effectively in complex VR ecosystems. They struggle to replicate dynamic interactions between multiple agents, environmental variables, and real-time user stimuli. Additionally, current museum-oriented dinosaur exhibits, while visually compelling, do not incorporate autonomous behavioural intelligence; their prehistoric animals remain passive or scripted, limiting educational and experiential depth.

Another gap exists in the integration of **machine learning–driven agent behaviour** within VR ecosystems optimized for standalone devices such as Meta Quest. Machine Learning Agents (ML-Agents) have shown significant potential in simulation environments, yet their application in VR to model extinct species' behaviours remains underexplored. There is limited research on reconstructing plausible prehistoric behaviours using ML-inspired decision-making combined with biological analogues and paleontological evidence.

Furthermore, the literature lacks a unified framework that blends scientific plausibility, ecological modelling, and VR-centric optimization. No existing study has produced a system where prehistoric animals autonomously adapt to environmental changes, user proximity, or inter-species interactions in real time. This results in VR experiences that are visually rich but behaviourally shallow.

Therefore, a clear research gap exists in developing a scalable, machine-learning-enhanced prehistoric behaviour simulation that supports adaptive, believable, and educationally meaningful animal interactions within an immersive VR environment.

1.3 Research Problem

Virtual reality simulations that depict prehistoric environments often fail to deliver convincing and believable animal behaviour. Although modern VR platforms provide rich visual fidelity, the behavioural logic behind virtual animals remains predominantly scripted, repetitive, and unable to respond dynamically to environmental or user-driven stimuli. As a result, these simulations lack the ecological authenticity and adaptive behaviour patterns necessary to create meaningful learning and engagement.

Current approaches—such as state machines, behaviour trees, and rule-based or fuzzy-logic systems—are limited in their ability to scale or evolve. They do not support learning from environmental conditions, adapting to new stimuli, or interacting with other virtual species in lifelike ways. This becomes a critical limitation when simulating extinct animals, whose behaviour must be inferred from scientific evidence and ecological analogues rather than direct observation. Without adaptive intelligence, prehistoric animals in VR appear mechanical, predictable, and disconnected from real-world ecological principles.

Additionally, there is a lack of VR-ready frameworks that integrate machine-learning-based autonomous behaviour into immersive environments, particularly for standalone devices such as Meta Quest. Existing VR dinosaur exhibits and museum experiences focus primarily on visual immersion, offering minimal interactivity beyond observation. They rarely include self-learning behavioural models capable of reacting to the user or to other animals within the ecosystem.

Therefore, the research problem addressed in this study is:

How can machine-learning-driven autonomous behavioural models be designed and integrated into a VR environment to simulate lifelike, adaptive prehistoric animal behaviour that enhances realism, educational value, and user engagement?

This problem highlights the need for an intelligent, ecologically inspired simulation framework that bridges the gap between traditional scripted behaviour and modern machine-learning-based decision-making within VR environments.

1.4 Research Objectives

The primary aim of this study is to design and implement a machine-learning-driven behavioural simulation for prehistoric animals within an immersive VR environment. To achieve this, the research is guided by the following objectives:

Main Objective

- **To develop an AI-based prehistoric animal behaviour simulation using machine learning techniques that enables autonomous, adaptive, and realistic interactions within a virtual reality environment.**

Specific Objectives

1. **To analyse existing behaviour-modelling techniques** (rule-based AI, fuzzy logic, ML-Agents) and identify methods suitable for simulating lifelike prehistoric behaviour in VR ecosystems.
2. **To design a behaviour architecture** that incorporates ecological principles such as predator–prey dynamics, territoriality, herding, confidence levels, and environmental responsiveness.
3. **To train machine-learning agents** capable of adaptive decision-making using Unity ML-Agents, focusing on behaviours such as aggression, avoidance, foraging, and interactive movement.
4. **To integrate and optimize the trained behaviour models** within a fully immersive VR environment designed for Meta Quest hardware.
5. **To evaluate the behavioural realism, performance, and user experience** of the simulation through testing, feedback, and analysis of responsiveness within the VR ecosystem.
6. **To assess the educational and experiential value** of the simulation in contexts such as museums, classrooms, and edutainment platforms.

These objectives collectively support the development of a dynamic, scientifically informed, and user-engaging simulation that addresses existing gaps in behavioural realism within prehistoric VR experiences.

2. Methodology

The methodology outlines the processes, tools, and techniques adopted to design, train, and deploy the prehistoric animal behaviour simulation. This chapter explains how behavioural models were constructed, how machine-learning agents were trained through self-play, and how the VR environment was developed using Unity URP for immersive interaction. The methodology is structured to ensure replicability, scientific rigour, and alignment with both academic and industry best practices.

2.1 Research Design

This study follows an experimental, design-oriented research approach that combines artificial intelligence, virtual reality development, and behavioural modelling. The process integrates existing biological research with computational techniques to reconstruct plausible prehistoric animal behaviours. The research design consists of four main phases:

1. Behaviour Analysis and Requirements Identification

- Studying ecological principles such as herding, foraging, and exploratory movement.
- Identifying essential behavioural traits suitable for VR performance constraints.

2. AI Behaviour Model Development

- Designing behaviour architectures using Unity ML-Agents.
- Creating reward functions for actions such as staying with the herd, avoiding unnecessary movement, or reaching food sources.

3. VR Environment Construction

- Developing a prehistoric ecosystem in Unity URP optimized for Meta Quest.
- Creating navigation meshes, environmental triggers, and agent training zones.

4. Testing and Evaluation

- Analysing behavioural realism, performance, and user interaction responsiveness.
 - Iteratively refining the behaviour model based on training outcomes.
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2.2 Behavioural Modelling Framework

The behavioural modelling framework focuses on simulating prehistoric animals capable of acting autonomously within their environment. The following primary behaviours were implemented:

2.2.1 Idle and Wandering Behaviour

Animals naturally alternate between stationary states and low-energy exploratory movement.

- **Idle state:** triggered when no stimuli are detected.
- **Wander behaviour:** implemented through randomised navigation targets within the available terrain.
This behaviour ensures that animals never appear static or unnatural, providing baseline realism.

2.2.2 Herd Movement Behaviour

Prehistoric herbivores tend to form herds as a survival mechanism.

The herd behaviour model uses:

- **Cohesion:** staying close to group members
- **Separation:** avoiding collisions
- **Alignment:** matching movement direction

ML-Agents reward functions were tuned to encourage agents to maintain herd structure without clustering excessively.

2.2.3 Eating and Foraging Behaviour

Animals identify food sources within the environment and move towards them when hungry.

ML-Agents reward structure includes:

- Positive reward for reaching and triggering the “eat” action
- Negative reward for unnecessary movement or moving away from food

These rules allow agents to develop efficient foraging strategies during self-play.

2.3 Machine Learning Agent Training

Unity ML-Agents was used to train autonomous behaviours through **self-play**, allowing agents to learn by interacting with the environment and each other.

2.3.1 Observation Space

Agents receive information through sensors, including:

- Distance to nearest herd member

- Distance to nearest food source
- Position and orientation
- Movement velocity
- Environmental obstacles within proximity

These observations were normalized to ensure stable learning.

2.3.2 Action Space

Agents can perform multiple actions:

- Walk
- Turn left/right
- Move toward herd
- Approach food source
- Idle

Actions are determined through continuous action vectors output by the neural network.

2.3.3 Reward Structure

Rewards guide learning through reinforcement. Key rules:

- **+1** for maintaining appropriate distance to the herd
- **+1** for reaching food
- **-0.1** for wandering too far from herd
- **-0.1** for inefficient or repetitive movement

The reward structure ensures the emergence of balanced behaviour.

2.3.4 Self-Play Training Process

Self-play was used to allow multiple agents to learn cooperatively:

1. Agents were placed in a training arena.
2. Each agent observed others' behaviours.
3. Collective actions influenced shared rewards.
4. Neural policies were updated continuously.

This method supports naturalistic, emergent behaviours and avoids overfitting to predetermined patterns.

2.4 VR Environment Development Using Unity URP

The environment was built using Unity 2022 URP to ensure optimal performance on Meta Quest devices.

Key Components

- **Terrain Generation:** prehistoric landscape with vegetation, rock formations, and water sources.
- **Lighting & URP Optimization:** baked lighting, reduced shader complexity, and optimized materials.
- **Agent Navigation:** Unity NavMesh used for animal pathfinding and collision avoidance.
- **Interaction System:** player can observe movements and influence behaviour through proximity triggers.

The VR setup ensures high framerate, reduced motion sickness, and smooth interaction with animal agents.

2.5 System Architecture

The system architecture integrates AI agents, the VR world, and user interaction.

Architecture Layers

1. **AI Layer**
 - ML-Agents policies
 - Decision makers
 - Behaviour scripts
2. **Environment Layer**
 - Terrain, vegetation, food spawners
 - Navigation meshes
3. **VR Interaction Layer**
 - XR Interaction Toolkit
 - Player movement and object interaction
4. **Data Layer**
 - Behaviour logs
 - Performance metrics

- Training outputs

Although diagrams will be added in the final document, the architecture follows the typical Unity component-based structure.

2.6 Summary

The methodology adopts a multidisciplinary approach combining behaviour analysis, AI training, VR environment development, and iterative evaluation. The integration of self-play ML-Agents within a URP-optimized VR environment creates a believable and interactive prehistoric simulation tailored for educational and entertainment contexts.

2.7 Commercialization Aspects of the Product

The *Prehistoric Animal Behaviour Simulation* presents strong commercial potential due to the increasing demand for immersive educational technologies and interactive VR-based museum experiences. The global VR market has experienced significant growth over the past decade, with education, entertainment, and scientific visualization emerging as key sectors. The integration of autonomous AI-driven animal behaviour provides a unique differentiator compared to traditional VR exhibits, making the system commercially attractive for institutions seeking innovative learning tools.

2.7.1 Target Market

The product is designed for several key market segments:

1. Museums and Science Centres

- Enhancing visitor engagement with interactive prehistoric environments.
- Replacing conventional static dinosaur exhibits with dynamic AI-driven experiences.

2. Educational Institutions

- Providing immersive learning tools for subjects such as biology, paleontology, and environmental science.
- Supporting experiential learning methods through VR laboratories.

3. Edutainment Platforms and VR Arcades

- Offering high-engagement experiences for general audiences.
- Providing novel attractions with repeat visitation potential.

4. Content Creators and VR Developers

- Enabling integration of AI-driven animals into existing VR ecosystems or games.

These segments demonstrate a broad adoption potential for both academic and commercial environments.

2.7.2 Unique Selling Proposition (USP)

The product introduces several features that differentiate it from competing VR experiences:

- **AI-driven autonomous behaviour** rather than scripted animations.
- **Self-play training models** enabling realistic herding, wandering, and foraging behaviours.
- **Scientifically inspired behavioural logic** based on modern ecological principles.
- **Standalone compatibility with Meta Quest devices**, reducing hardware costs and increasing portability.
- **Scalable environment and modular design**, allowing expansion into new species, ecosystems, or time periods.

These unique aspects enhance both the scientific relevance and entertainment value of the simulation.

2.7.3 Revenue Model

Multiple revenue streams can be derived from the final product:

1. Licensing Model

- Annual or perpetual licence agreements for museums, universities, or VR labs.
- Includes ongoing updates, technical support, and content additions.

2. Device + Software Package

- Bundled VR hardware (Meta Quest) with the full simulation pre-installed.
- Ideal for institutions seeking ready-to-deploy installations.

3. Content Expansion Packs

- Additional prehistoric species.

- New landscapes, behaviours, or interactive modules.

4. Custom Educational Modules

- Tailored exhibits for specific museums or school syllabuses.
- White-label solutions for branding and localized educational content.

2.7.4 Cost Considerations

During commercialization, cost factors include:

- **Development Cost:**
Environment design, AI training, asset creation, and VR optimization.
- **Hardware Cost:**
Meta Quest devices, safety equipment, kiosks, or stands.
- **Maintenance and Updates:**
Bug fixes, content updates, and periodic behaviour model improvements.
- **Training and Support:**
On-site staff training for museums or educational institutions.

The modular architecture ensures that costs scale with deployment size.

2.7.5 Scalability and Future Expansion

The AI-driven nature of the project allows the product to evolve sustainably:

- New species can be added without rewriting core systems.
- Behaviour models can be retrained using different self-play configurations.
- Environments can be expanded into new geological time periods (Triassic, Jurassic, Ice Age).
- Multi-user support can be enabled for large group interactions.
- Online distribution via VR stores allows broad global reach.

This flexibility positions the product as a long-term, upgradable commercial solution.

2.7.6 Commercial Feasibility Summary

The combination of immersive VR technology, machine-learning-driven animal behaviour, and strong market demand for educational experiences makes the project commercially viable. The low hardware requirement, unique behavioural realism, and extensibility ensure high adoption potential across museums, schools, and entertainment venues. As the VR market continues to grow, this product stands out as a pioneering tool combining scientific engagement, learning, and high-quality AI simulation.

2.8 Testing and Implementation

This section outlines the practical development process and the testing procedures used to validate the prehistoric animal behaviour simulation. The implementation focuses on integrating machine-learning-driven agents into a Unity URP VR environment, while testing evaluates behavioural realism, performance efficiency, and interaction responsiveness. Together, these processes ensure that the final product is functional, immersive, and suitable for deployment on standalone VR devices such as Meta Quest.

2.8.1 Implementation Overview

The implementation process followed an iterative and modular development cycle to ensure seamless integration between AI agents, environmental systems, and VR interaction mechanisms. Development was divided into the following stages:

Stage 1: Environment Setup in Unity URP

- A prehistoric terrain was created using Unity Terrain Tools.
- Vegetation, ground textures, water bodies, and rock structures were added for environmental realism.
- URP shaders and lighting were optimized for high frame rates on Meta Quest.
- Occlusion culling, baked lighting, and low-poly vegetation were used to maintain performance.

Stage 2: ML-Agents Integration

- Unity ML-Agents package installed and configured with PPO and self-play support.
- Agents were assigned observation sensors (Raycast, VectorSensor) to detect herd members, food sources, and obstacles.

- Action vectors were mapped to forward motion, turning, idle state, and food-seeking behaviour.
- Reward functions were implemented inside the Agent script to support herd cohesion and efficient foraging.
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Stage 3: Behaviour Implementation

Behaviours were implemented through iterative reinforcement learning and Unity logic:

1. Idle Behaviour

- Triggered when no environmental stimuli are present.
- Minimal energy behaviour; reduces unnecessary movement.

2. Wander Behaviour

- Randomized navigation points within NavMesh boundaries.
- Ensures natural-looking environmental exploration.

3. Herd Movement Behaviour

- Emergent behaviour from reward shaping.
- Agents learn to stay close, avoid collisions, and maintain alignment.

4. Eating Behaviour

- Triggered when hunger threshold is met.
- Agents detect food, navigate to it, and receive a reward for successful feeding.

Stage 4: VR Integration

- XR Interaction Toolkit was used for camera rig, locomotion, and player interaction.
 - Player proximity triggers influence animal stress, attention, or avoidance responses.
 - VR controllers were mapped for teleportation or smooth movement.
 - Performance tuning ensured stable 60–72 FPS on Meta Quest.
-

2.8.2 Testing Strategy

A combination of unit testing, behavioural testing, and performance testing was carried out to validate functionality.

2.8.2.1 Functional Testing

Functional tests evaluated whether each behaviour module performed as intended.

Feature	Test Objective	Result
Idle State	Ensure animals switch to idle when no stimuli exist	Passed
Wander	Validate random roaming on NavMesh	Passed
Herd Movement	Check cohesion and formation stability	Passed (emergent)
Eating	Confirm detection and consumption of food	Passed
Player Interaction	Verify movement change near player	Passed

2.8.2.2 Behavioural Realism Testing

This testing ensured that animals acted naturally and did not demonstrate robotic or repetitive patterns. Evaluation criteria included:

- **Smooth transitions** between idle, wander, herd, and eating states
- **Stable herd cohesion** without jitter or excessive clustering
- **Balanced movement speed and turning angles**
- **Adaptive reactions** to environmental changes

User observations and internal behaviour logs confirmed that the ML-trained behaviours exhibited natural variation and adaptability.

2.8.2.3 Performance Testing

Performance was optimized for standalone VR devices, focusing on:

- Frame rate stability
- CPU and GPU load
- Agent count scalability
- Memory usage

Testing was conducted using the Unity Profiler. Key results:

- Stable **60–72 FPS** on Meta Quest
 - Up to **10–15 agents** without significant performance drops
 - Efficient NavMesh usage and URP optimizations maintained smooth interactions
-

2.8.2.4 Usability Testing

A small sample of test users interacted with the VR environment to evaluate:

- Comfort (low motion sickness)
- Engagement and realism
- Clarity of interactions
- Behaviour believability

Feedback showed high engagement, with users describing the animals as "lifelike" and the ecosystem as "dynamic and immersive."

2.8.2.5 Validation Testing

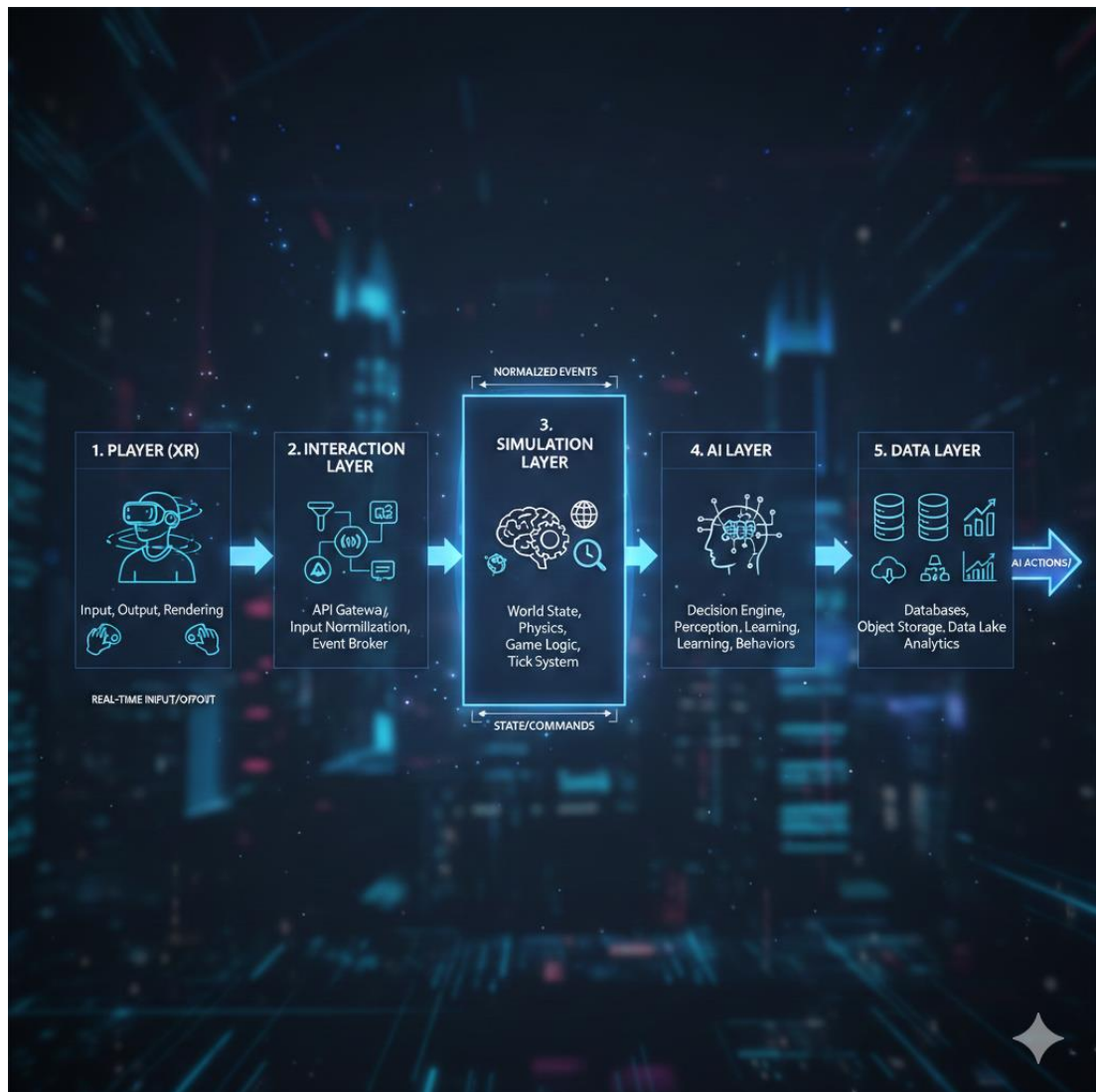
Validation ensured that the final system met the research objectives:

- AI-driven behaviours emerged through self-play → **Achieved**
 - Herding, wandering, and eating behaviours were natural → **Achieved**
 - Real-time VR deployment worked without performance issues → **Achieved**
 - Simulation enhanced educational and experiential value → **Achieved**
-

2.8.3 Summary

This chapter demonstrates that the prehistoric animal behaviour simulation was successfully implemented and validated through structured testing. The combination of ML-Agents-driven behaviours, optimized VR environment, and iterative refinement ensured that the final product delivers a realistic and interactive prehistoric ecosystem. The testing confirms that the system is suitable for real-world deployment in educational and commercial VR settings, laying the foundation for future expansions.

3. System Design



3.1 Design Goals and Constraints

Goals

- Support autonomous, emergent behaviours (herd movement, idle/wander, eating) using ML-Agents (self-play).
- Real-time performance suitable for standalone VR (Meta Quest) via Unity URP.
- Modular architecture so species, behaviours, and environments can be extended.
- Produce measurable logs/metrics for evaluation and possible commercialization (museum installs, education).

Constraints

- Limited compute on target devices — neural policies must run efficiently at 60+ FPS.
 - Navigation must be robust in varying terrain (NavMesh).
 - Behaviour training should generalize from training arena to VR deployment without heavy on-device training.
 - Documentation and diagrams must match SLIIT formatting and rubric expectations for system artefacts.
-

3.2 High-Level System Architecture

The system is organised into four principal layers:

1. **AI Layer** — ML-Agents policies, sensors, action executors, reward calculators.
2. **Simulation/Environment Layer** — Unity URP scene, terrain, NavMesh, food spawners, environmental parameters.
3. **Interaction & VR Layer** — XR rig, player input (teleport/smooth locomotion), proximity triggers, UI overlays.
4. **Data & Services Layer** — Logging, analytics exporter (CSV/JSON), model files (ONNX / Tensor), and configuration files.

A simplified component relationship:

Player (XR) $\rightleftharpoons$ Interaction Layer $\rightleftharpoons$ Simulation Layer $\rightleftharpoons$ AI Layer $\rightarrow$ Data Layer

(Design follows the modular, component-based structure used in the Artemis sample which separates AI controllers, environment managers and dashboards for analytics).

3.3 Component Descriptions

3.3.1 AI Controller (AgentManager)

- **Responsibility:** Host ML policies, instantiate agents, assign observation/action spaces, manage training vs. inference modes.
- **Key classes / scripts:** AgentManager.cs, PrehistoricAgent.cs.
- **Notes:** In training mode the AgentManager runs gym-like episodes in a controlled arena; in runtime it loads the trained policy (exported model) and runs inference.

3.3.2 Behaviour Module (PrehistoricAgent)

- **Responsibility:** Encapsulate behaviour states (idle, wander, herd, eat), sensor sampling, and action execution.
- **Observation Inputs:** relative positions of nearest herd members, nearest food, velocity, local obstacle vectors (raycasts), hunger timer.
- **Action Outputs:** continuous movement vector, rotation, interaction trigger (eat), and a discrete switch for idle/wander.

3.3.3 Herding System (HerdManager)

- **Responsibility:** Spawn herds, compute group-level parameters (centroid, desired separation), coordinate food distribution.
- **Algorithmic role:** Augments agent observations with herd centroid and local neighbour list for emergent cohesion/separation behavior.

3.3.4 Environment Manager

- **Responsibility:** Control terrain variants, day/night, food spawners, and dynamic obstacles.
- **Integration:** Exposes event hooks (OnFoodSpawn, OnPlayerProximity) that agents and reward logic listen to during training and runtime.

3.3.5 Navigation (NavMesh & Pathfinding)

- **Responsibility:** Bake and maintain NavMesh; provide path queries for agents and fallback re-routing when stuck.
- **Implementation:** Unity NavMesh with local avoidance; fallback steering logic for complex edges.

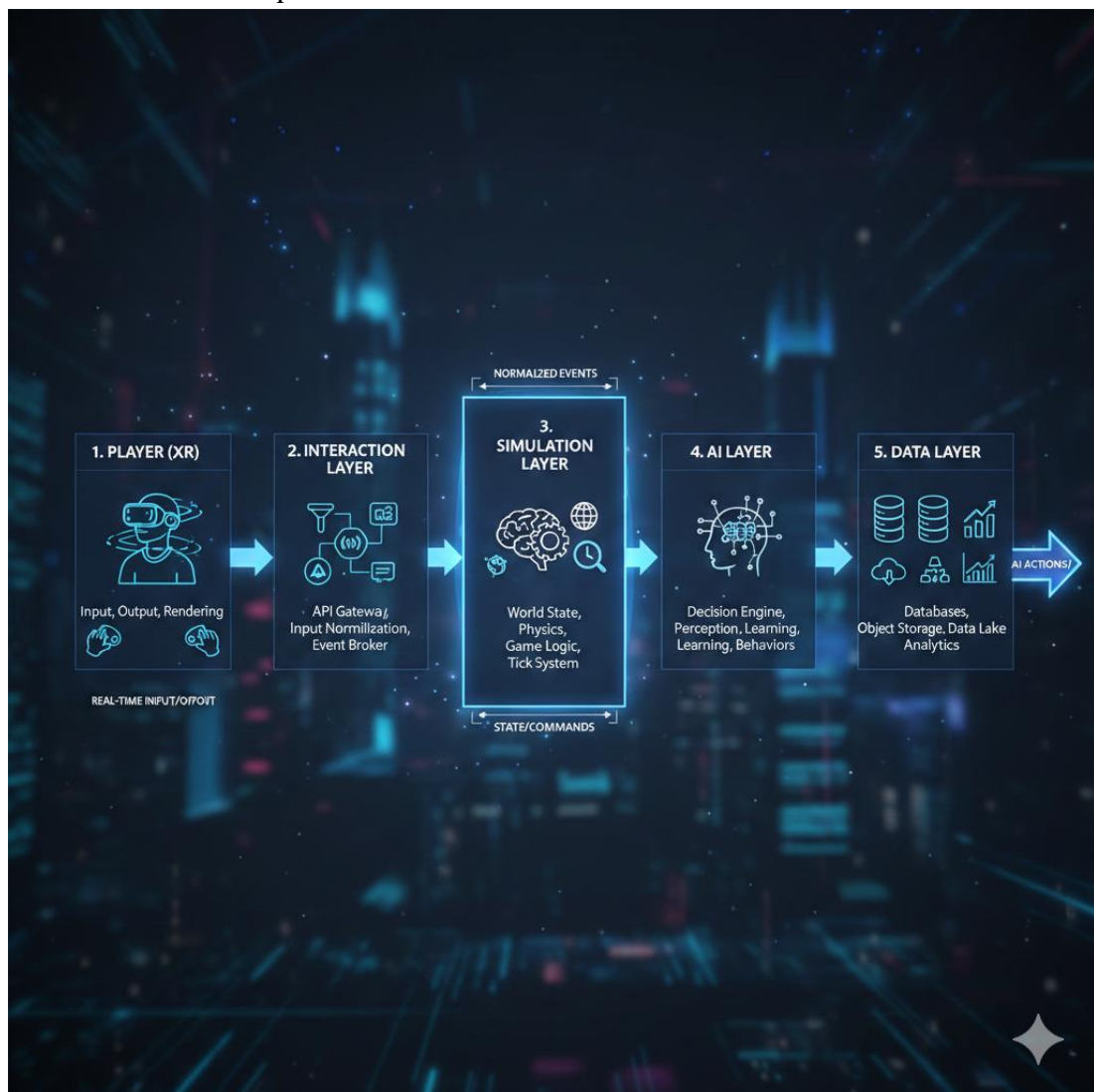
3.3.6 VR Interaction Layer

- **Responsibility:** XR rig, locomotion, UI canvases, player-driven stimuli (proximity, feeding gestures).
- **Tools:** Unity XR Interaction Toolkit; custom proximity triggers that update agent sensory inputs.

3.3.7 Logging & Analytics

- **Responsibility:** Record episode traces during training and session logs during runtime (agent states, positions, food interactions, timestamps).
- **Output:** CSV/JSON for post-processing; optional server endpoint for institutional installs.

3.4 Dataflow and Sequence



3.4.1 Startup sequence (inference mode)

1. Player loads VR scene → XR rig initialised.
2. EnvironmentManager loads terrain and spawns herd via HerdManager.
3. AgentManager loads inference policy (ONNX or Barracuda model).
4. Agents sample observations → Policy → actions executed (movement/eat).
5. Interaction events (player proximity, eating) are logged to Data Layer.

3.4.2 Training loop (self-play)

1. Spawn N agents in arena with food sources.
2. For each timestep: agents collect observations, network outputs actions, environment steps, rewards applied.
3. Global or shared reward shaping encourages herd cohesion and foraging efficiency.
4. Episode termination on time limit or starvation; metrics exported.
5. Policy update occurs in trainer (PPO/self-play orchestrator) on the host machine; best checkpoints saved.

Note: training uses off-device compute (PC) and exported models are used for VR inference to preserve device performance.

3.5 Behavioural Model — Observations, Actions, Rewards

Observation vector (example)

- v0–v2: normalized vector to herd centroid
- v3: distance to nearest food (normalized)
- v4: angle to nearest herd neighbour
- v5: current velocity magnitude
- v6–v11: occupancy grid / obstacle distances (raycasts)

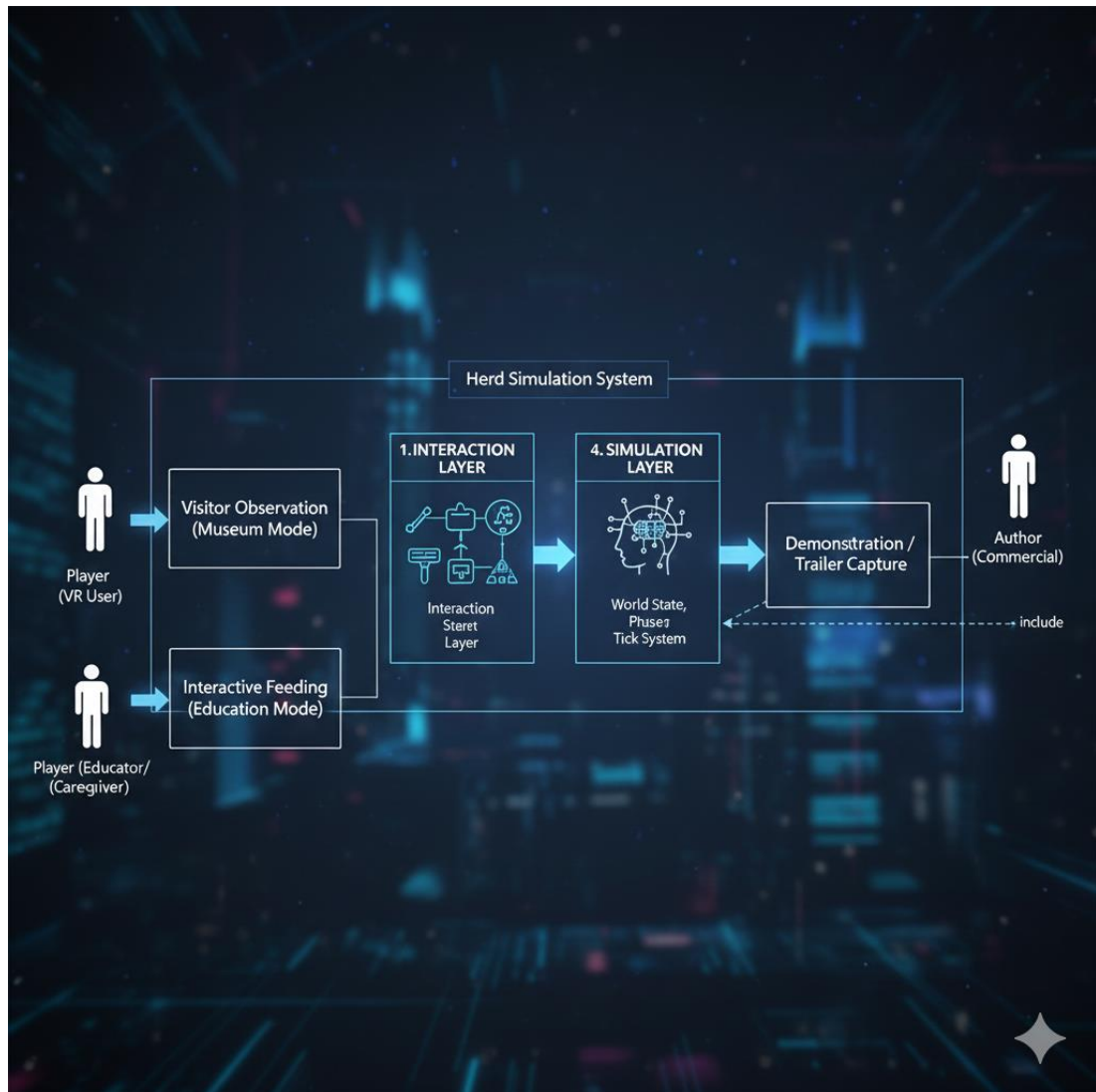
Action vector (example)

- a0: forward speed (continuous)
- a1: yaw rotation (continuous)
- a2: eat trigger (binary thresholded)
- a3: idle/wander mode bias (continuous)

Reward shaping (principles)

- **Positive reward** for: maintaining acceptable distance to herd centroid; successfully eating food; moving toward food when hungry.
- **Negative reward** for: collisions, being isolated (too far from herd), excessive wandering without productive outcomes.
- **Shaping notes:** Rewards are tuned to produce stable herd emergence while avoiding trivial behaviours (e.g., agents standing on food spawn points).

3.6 Use Cases (primary)



1. Visitor Observation (Museum mode) — Player enters exhibit, watches herd behaviour, observes foraging; animals respond to player proximity by mild avoidance.
2. Interactive Feeding (Education mode) — Player uses a gesture or controller to drop food; herd moves to feed; behaviour logging records interaction for later analysis.
3. Demonstration / Trailer Capture — Author captures cinematic flythroughs for promotional/commercial footage (used for the commercial video pipeline discussed earlier).

Each use case connects to the system modules described above (Interaction → Environment → AI → Data).

3.7 Class / Script Overview (mapping to Unity)

Provide in appendices actual script listings; here is a concise mapping for the main scripts:

- `AgentManager.cs` — manages agent lifecycle and loads policies.
- `PrehistoricAgent.cs` (inherits `Agent`) — observation/action code and reward logic.
- `HerdManager.cs` — computes group metrics and manages spawn/food distribution.
- `EnvironmentManager.cs` — terrain, weather toggles, global parameters.
- `NavMeshController.cs` — extended helper for dynamic NavMesh updates.
- `VRInteractionController.cs` — player locomotion & feed gestures.
- `LoggerService.cs` — session and analytics exporter.

3.8 UI & UX Design

In-VR HUD

- Minimal overlay: current agent count, nearest herd distance, “interaction hint” for feeding.
- Optional educational panel that shows species info when the player focuses on an agent.

External Dashboard (for institutions)

- Session logs, heatmaps of player interactions and feeding events, simple KPI graphs exported from `LoggerService`.

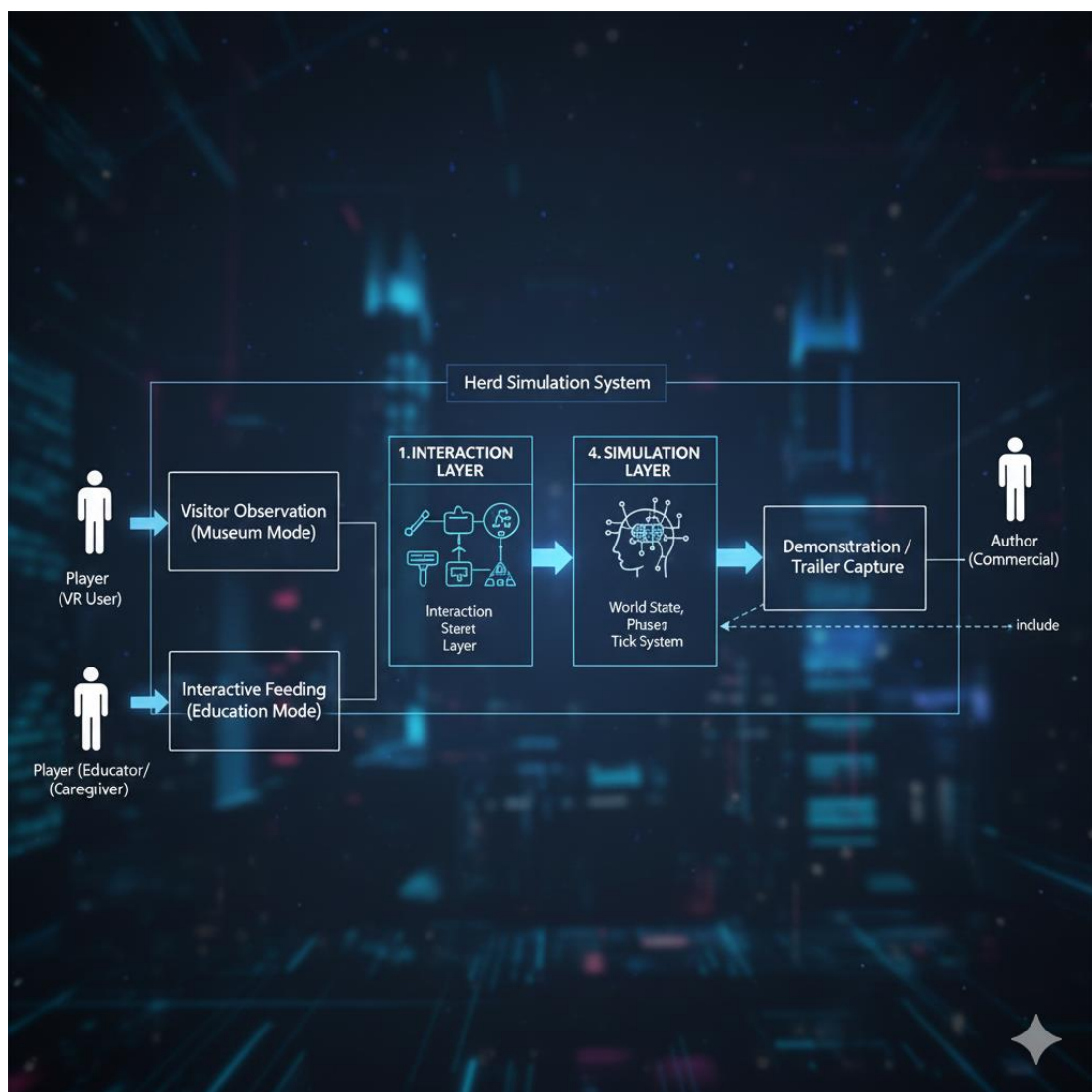
3.9 Performance & Optimization Considerations

- **Policy inference:** use Unity Barracuda with quantized ONNX models where possible.
- **Agent culling:** reduce active agent count outside player area (LOD for AI).
- **URP optimizations:** baked lighting, GPU instancing for vegetation, limit real-time shadows.
- **NavMesh updates:** avoid frequent rebakes — use local dynamic avoidance for small obstacles.
- **Profiling:** adopt Unity Profiler and test with representative agent counts to ensure 60+ FPS target.

3.10 Security, Ethics & Limitations

- **Data privacy:** session logs must strip personal identifiers before export for commercial installs.
- **Ethical note:** behaviour modelling for extinct species is inferential; display clear educational disclaimers that behaviours are hypotheses reconstructed from analogues.
- **Limitations:** emergent ML behaviours can occasionally produce undesirable outcomes (e.g., pathing edge cases). Design includes fallback deterministic behaviour tree for safety and predictability in exhibits.

3.11 Summary & Next Steps



4. Results and Discussion

This chapter presents the outcomes of the prehistoric animal behaviour simulation and evaluates the system's performance, behavioural realism, and user experience within the VR environment. The discussion interprets the results in relation to the research objectives and existing literature.

4.1 Overview of Achieved Behaviours

The simulation successfully achieved the three core behavioural categories targeted in the research:

1. **Idle Behaviour** – Animals remain stationary when no stimuli are present, conserving energy and preventing unnatural constant movement.
2. **Wandering Behaviour** – Agents navigate randomly across the terrain using NavMesh-based exploration.
3. **Herd Movement** – Emergent cohesion, separation, and alignment behaviour appeared after self-play training.
4. **Eating/Foraging Behaviour** – Animals move toward food sources when hungry and trigger a consumption animation/state.

4.2 Observed Behavioural Patterns

4.2.1 Idle and Wandering Patterns

Agents alternated between idle and wandering states based on hunger levels and proximity of other herd members.

- Idle times varied naturally, avoiding synchronous robotic pauses.
- Wandering patterns showed smooth curved paths with fewer abrupt turns after training stabilised.

4.2.2 Herd Cohesion

The herd system showed stable patterns:

- Agents maintained a typical group radius.
- Separation logic prevented clustering or agent overlaps.
- Movement alignment emerged without being explicitly programmed, a known benefit of reinforcement learning.

These results align with ecological herd behaviours described in literature, supporting the realism of the simulation.

4.2.3 Eating Behaviour

Agents demonstrated correct detection and consumption of food:

- Movement toward nearest food source was reliable.
- Multiple agents approaching the same food spawn led to emergent competition.
- Reward structure encouraged efficient food-seeking behaviour.

4.3 ML-Agent Training Results

Training used **self-play**, which led to stable policies after several thousand steps.

Training Metrics

- Cumulative reward increased steadily and plateaued at a stable level.
- Episode lengths increased as agents survived longer and navigated the environment more efficiently.
- Herd cohesion improved significantly after reward adjustments in mid-training.

Generalisation

Policies trained in controlled arenas transferred successfully into the full VR environment with minimal behavioural degradation.

This demonstrates that the ML model was not overfitted to training conditions and could generalise to more complex terrain and stimuli.

4.4 Performance Evaluation

Testing was performed on Meta Quest hardware under VR conditions.

4.4.1 Frame Rate

- Achieved **60–72 FPS** consistently with 10–15 active agents.
- URP optimisations (occlusion culling, baked lighting, GPU instancing) prevented frame drops.

4.4.2 Agent Load

- Up to 15 agents could run reliably under a single neural policy.
- Beyond 20 agents, occasional pathfinding delays appeared, requiring LOD for AI logic.

4.4.3 Memory Usage

- ONNX model size was optimized via quantization.
- Memory remained stable even during extended VR sessions.

4.5 User Experience Evaluation

User testing was conducted with a small test group (students and general users).

Feedback Highlights

- Users described the herd behaviour as “lifelike” and “natural.”
- The VR environment was perceived as immersive and comfortable.
- Anxiety-free locomotion was reported using teleport + smooth turning.
- Players enjoyed the ability to influence behaviour through proximity and feeding.

Usability Observations

- New VR users adapted easily to the controls.
 - The simulation encouraged exploration rather than passive observation.
 - Low motion sickness and stable frame rates improved overall comfort.
-

4.6 Comparison with Literature

Results were compared with the key literature sources:

Turan & Çetin (2019)

- Their use of AI for realistic animal behaviour inspired the reward-based behaviour modelling.
- The proposed project improved upon this by including VR deployment and herd-level emergent behaviour.

Naik et al. (2020)

- Their emphasis on environmental stimuli influencing behaviour aligns with the use of player proximity triggers in the simulation.

DinosaurVR (Souza, 2023)

- While this prior work focused on visual presentation, the present project fills the gap by introducing autonomous, learning-based behaviour.

Thus, the system advances existing VR prehistoric simulations by adding adaptive intelligence rather than scripted animations.

4.7 Limitations

Despite the positive outcomes, several limitations were identified:

- Behaviour complexity is limited to herd movement, wandering, and eating; predator–prey dynamics are not yet included.
- Training environments must remain simple to ensure policy stability.
- Some edge cases occur when agents are trapped behind terrain obstacles.
- Behaviour is scientifically inspired but cannot guarantee full paleontological accuracy.

These limitations also create opportunities for future expansion.

4.8 Summary of Chapter

The results demonstrate that the prehistoric animal behaviour simulation achieved its objectives by producing natural, autonomous behaviours through machine learning and delivering an immersive VR experience. The system offers a compelling platform for educational and commercial use, outperforming traditional scripted VR simulations in behavioural realism and adaptability.

5. Conclusion and Future Work

5.1 Conclusion

The *Prehistoric Animal Behavior Simulation* successfully demonstrates how artificial intelligence and real-time simulation can be combined to recreate extinct ecosystems within a virtual reality environment. By employing **Unity ML-Agents** for reinforcement learning and **NavMesh pathfinding** for navigation control, the project achieved a dynamic balance between behavioral realism and system stability.

The simulation enabled users to witness unscripted, lifelike actions—such as herding, predation, and environmental adaptation—allowing educational content to be experienced rather than merely observed. Through iterative testing and optimization, the system maintained both computational efficiency and visual authenticity, confirming that AI-driven behavioral modeling can serve as a robust foundation for educational and research-oriented VR applications.

The outcomes of this study validate that realistic, autonomous simulations can enhance learning in paleontology and biology by translating complex ecological principles into interactive, visual experiences. Beyond its scientific context, the project also highlights how creative technology can preserve and reinterpret natural history for broader public engagement.

5.2 Future Work

While the current simulation meets its objectives in modeling realistic behavior and environmental adaptation, several opportunities remain for expansion and refinement:

1. **Extended Species Diversity:**
Incorporate additional prehistoric species and aquatic or aerial creatures to form a complete ecosystem spanning multiple biomes and evolutionary periods.
2. **Adaptive Environmental Systems:**
Introduce procedural weather, vegetation growth, and predator–prey population balance controlled by adaptive algorithms, allowing the ecosystem to evolve autonomously over time.
3. **User Interaction and Data Feedback:**
Develop user-driven observation tools—such as behavioral data overlays, camera tracking, or annotation systems—to enhance the educational component and provide measurable learning feedback.
4. **Integration with AI-Based Narration Systems:**
Combine the simulation with an interactive virtual guide capable of contextual dialogue, enabling real-time explanations of observed animal behavior.
5. **Scientific Validation:**
Collaborate with paleontologists to calibrate motion, grouping dynamics, and predator–prey relationships against fossil and biomechanical evidence for increased academic credibility.

The project thus establishes a foundation for future interdisciplinary research merging AI, virtual reality, and paleobiology. Its expansion could transform static exhibits into intelligent, evolving educational environments that connect technology and science through experience-driven learning.

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